Hints: Limits – 3. Asymptotics of sequences

Class work

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. The sequence consists of blocks $1, 2, \dots, m$. Compute the sum $S_n = \sum_{k \leqslant n} a_k$ exactly when $n = T_m = \frac{m(m+1)}{2}$ is the end of the m-th block.

Hint 2. $S_{T_m} = \sum_{j=1}^{m} \frac{j(j+1)}{2} = \frac{m(m+1)(m+2)}{6}$. For $T_m \leqslant n < T_{m+1}$ one has $S_{T_m} \leqslant S_n \leqslant S_{T_{m+1}}$, and $m \sim \sqrt{2n}$.

Hint 3. So $S_n \sim \frac{m^3}{6} \sim \frac{(2n)^{3/2}}{6} = \frac{\sqrt{2}}{3}\, n^{3/2}$. The limit is finite and positive only for $\alpha = \frac{3}{2}$; answer: $(\alpha, \beta) = (\frac{3}{2}, \frac{\sqrt{2}}{3})$.

Problem 2

Hint 1. Compare $F(a_n)$ with $n = F(b_n)$. By the mean value theorem $F(a_{k+1}) - F(a_k) = \dfrac{f(\xi_k)}{f(a_k)}$ with $\xi_k \in (a_k, a_{k+1})$, and since f increases this lies in $\Big[1, \dfrac{f(a_{k+1})}{f(a_k)}\Big]$. Summing,

$$n + F(1) \leqslant F(a_n) \leqslant n + F(1) + \sum_{k<n} \frac{f(a_{k+1}) - f(a_k)}{f(a_k)}.$$

In particular $a_n > b_n$ and $a_n \to \infty$.

Hint 2. Show that $D_n = F(a_n) - F(b_n) = F(a_n) - n$ is $o\big(f(a_n)\big)$. This is Stolz in disguise: the increment of D_n is at most $\frac{f(a_{n+1}) - f(a_n)}{f(a_n)}$, i.e. at most $\frac{1}{f(a_n)} \to 0$ times the increment of $f(a_n)$. (Directly: given ε, choose K with $f(a_K) > \frac{2}{\varepsilon}$ and split the sum at K; the tail is at most $\frac{\varepsilon}{2}\big(f(a_n) - f(a_K)\big)$.)

Hint 3. Mean value theorem once more: $D_n \geqslant f(b_n)(a_n - b_n)$, and with $B = \sup f'$ also $f(a_n) \leqslant f(b_n) + B(a_n - b_n)$. So for large n

$$f(b_n)(a_n - b_n) \leqslant \varepsilon\big(f(b_n) + B(a_n - b_n)\big), \qquad \text{i.e.} \qquad a_n - b_n \leqslant \frac{\varepsilon f(b_n)}{f(b_n) - \varepsilon B} < 2\varepsilon,$$

because $f(b_n) \to \infty$.

Problem 3

Hint 1. By induction $x_n > 0$. Bernoulli's inequality $(1+t)^{1/n} \leqslant 1 + \frac{t}{n}$ gives $x_{n+1} \leqslant x_n$, so $x_n \leqslant x_2$ and $x_{n+1} \leqslant \sqrt[n]{1 + nx_2} - 1 \to 0$. Hence $\lim x_n = 0$.

Hint 2. Put $y_n = nx_n$. Then $x_{n+1} = \exp\big(\frac{\ln(1+y_n)}{n}\big) - 1$, and $\ln(1 + y_n) \leqslant \ln(1 + nx_2) = o(n)$, so $y_{n+1} = \frac{n+1}{n} \ln(1 + y_n)\,(1 + o(1))$. Thus for every $\varepsilon > 0$ eventually $y_{n+1} \leqslant (1 + \varepsilon) \ln(1 + y_n)$.

Hint 3. The function $g(y) = (1+\varepsilon)\ln(1+y)$ is increasing and concave with $g'(0) = 1+\varepsilon$, so it has a unique positive fixed point y_ε, and $y_\varepsilon \to 0$ as $\varepsilon \to 0$ (in fact $y_\varepsilon \approx 2\varepsilon$). Iterating, $\limsup y_n \leqslant y_\varepsilon$. Answer: $\lim x_n = 0$ and $\lim n x_n = 0$.

Problem 4

Hint 1. Put $y_n = x_{n+1}^3 = \frac{1}{n}\sum_{k\leqslant n}\sin^3 x_k$. Then $(n+1)y_{n+1} = ny_n + \sin^3 x_{n+1}$, i.e. $y_{n+1} - y_n = \dfrac{\sin^3 x_{n+1} - x_{n+1}^3}{n+1} < 0$. Deduce that $y_n \downarrow 0$ (if $y_n \to L > 0$, the decrements are $\geqslant c/n$ and sum to ∞).

Hint 2. Since $\sin^3 t - t^3 = -\frac{t^5}{2} + O(t^7)$, we get $y_{n+1} = y_n - \dfrac{y_n^{5/3}}{2(n+1)}(1+o(1))$. This is a neutral fixed point with an extra factor $\frac{1}{n}$: telescope in the power $y^{-2/3}$.

Hint 3. $y_{n+1}^{-2/3} - y_n^{-2/3} = \dfrac{1+o(1)}{3(n+1)}$, so by Stolz with $b_n = \ln n$: $y_n^{-2/3} \sim \frac{\ln n}{3}$. Hence $x_n^2 \sim \frac{3}{\ln n}$, i.e. $\lim x_n^2 \ln n = 3$.

Problem 5

Hint 1. Work with the means $m_n = \frac{x_1+\cdots+x_n}{n} > 0$, so that $x_{n+1} = \arctan m_n$. Then $(n+1)m_{n+1} = nm_n + \arctan m_n$, i.e.

$$m_{n+1} = m_n - \frac{m_n - \arctan m_n}{n+1}.$$

Hence m_n decreases; if $m_n \to L > 0$, the decrements are $\geqslant \frac{c}{n+1}$ and sum to ∞. So $m_n \to 0$.

Hint 2. $m - \arctan m = \frac{m^3}{3} + O(m^5)$, so $m_{n+1} = m_n - \frac{m_n^3}{3(n+1)}(1+o(1))$: the same situation as in Problem 4. Telescope in the power m^{-2}.

Hint 3. $\dfrac{1}{m_{n+1}^2} - \dfrac{1}{m_n^2} = \dfrac{2+o(1)}{3(n+1)}$, so by Stolz with $b_n = \ln n$ we get $m_n^{-2} \sim \frac{2}{3}\ln n$. Since $x_{n+1} = \arctan m_n \sim m_n$, $x_n^2 \ln n \to \frac{3}{2}$. Answer: $\lim x_n\sqrt{\ln n} = \sqrt{3/2}$.